

2023 Paper

Question 1

Write a Java class to simulate the popular puzzle game Sudoku. In classic Sudoku, the objective is to fill a 9×9 grid with digits so that each column, each row, and each of the nine 3×3 subgrids that compose the grid (also called “boxes”, “blocks”, or “regions”) contain all the digits from 1 to 9. The puzzle setter provides a partially completed grid, which for a well-posed puzzle has a single solution.

3		6	5		8	4		
5	2							
	8	7					3	1
		3		1			8	
9			8	6	3			5
	5			9		6		
1	3					2	5	
							7	4
		5	2		6	3		

Your class should input a starting Sudoku grid with numbers, where zero represents empty squares. This input could be from the standard input or an argument of a method. Calculate the solution using a recursive algorithm and print the result to standard output. An example input grid and output is given below:

```
Input:
grid = { {3, 0, 6, 5, 0, 8, 4, 0, 0},
         {5, 2, 0, 0, 0, 0, 0, 0, 0},
         {0, 8, 7, 0, 0, 0, 0, 3, 1},
         {0, 0, 3, 0, 1, 0, 0, 8, 0},
         {9, 0, 0, 8, 6, 3, 0, 0, 5},
         {0, 5, 0, 0, 9, 0, 6, 0, 0},
         {1, 3, 0, 0, 0, 0, 2, 5, 0},
         {0, 0, 0, 0, 0, 0, 0, 7, 4},
         {0, 0, 5, 2, 0, 6, 3, 0, 0} }
Output:
          3 1 6 5 7 8 4 9 2
          5 2 9 1 3 4 7 6 8
          4 8 7 6 2 9 5 3 1
          2 6 3 4 1 5 9 8 7
          9 7 4 8 6 3 1 2 5
          8 5 1 7 9 2 6 4 3
          1 3 8 9 4 7 2 5 6
          6 9 2 3 5 1 8 7 4
          7 4 5 2 8 6 3 1 9
```

Question 2

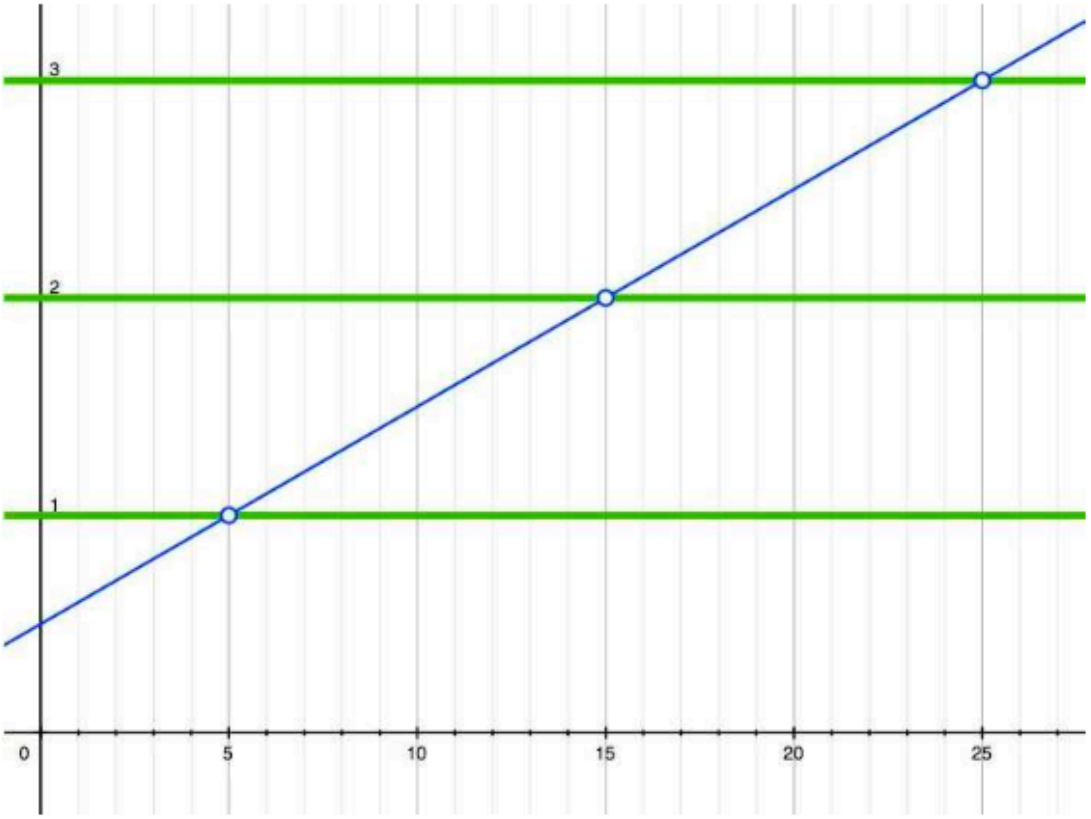
Three points with coordinates (x1, y1), (x2, y2), (x3, y3) are collinear when they satisfy the following equation:

$$\frac{y_3 - y_2}{x_3 - x_2} = \frac{y_2 - y_1}{x_2 - x_1}$$

Consider three unsorted integer arrays a1, a2, and a3 containing the x-coordinates of points on the horizontal lines y=1, y=2, and y=3, respectively (assume points are unique,
i.e. there are no duplicate numbers in the arrays). That is,

- a1[i] (for some i) represents the point (a1[i], 1);
- a2[i] represents the point (a2[i], 2); and
- a3[i] represents the point (a3[i], 3).

Below is an example of three collinear points.



We want to count how many sets of 3 points, one from each of the arrays a1, a2, and a3,
are **collinear,** i.e. lie on the same (non-horizontal) line.

1. Implement the following method in Java that counts this number.

   ```
   int countCollinear(int[] a1, int[] a2, int[] a3)
   ```

   Your implementation must run in O(N2), where N = a1.length = a2.length = a3.length. Your code can assume and call any known algorithms or data structures
   without implementing them. However you should name the algorithms or data structures you use and state their worst-case runtime cost.

2. Describe why your implementation correctly solves the problem.

3. Write a detailed analysis of your code that shows its runtime performance.